PROGRAMMING LABORATORY-ITIC 1152

Practical – 3

Re.Nu;-D/ICT/25/0121

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Question 1

Input;-

File Edit View

```
#include <iostream>
using namespace std;

int main() {
     cout<<"Yuwani Anuththara"<<endl;
     cout<< "Girl's High School,Mt. Lavinia";

     return 0;

}
```

Output;-

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER\OneDrive\Desktop\c++> g++ lb3(1).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Yuwani Anuththara
Girl's High School,Mt. Lavinia
C:\Users\USER\OneDrive\Desktop\c++>
```

Question 2

Input;-

File Edit View

```
#include <iostream>
using namespace std;

int main() {
    int intvalue = 56;
    float floatvalue = 56.651;
    double doublevalue = 56.65895;
    string stringvalue = "Hello";

    cout<<"Integer value :"<< intvalue<<endl;
    cout<<"Float value   :"<< floatvalue<<endl;
    cout<<"Double value  :"<< doublevalue<<endl;
    cout<<"String value  :"<< stringvalue<<endl;

    return 0;
}
```

Output;-

```
C:\Users\USER\OneDrive\Desktop\c++> g++ lb3(2).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Integer value :56
Float value   :56.651
Double value  :56.6589
String value  :Hello

C:\Users\USER\OneDrive\Desktop\c++>
```

Question 3

Input;-

File Edit View

```
#include <iostream>
using namespace std;

int main() {
    int num1 = 56, num2=25;
    int sum;
       sum=num1+ num2;
         cout<<"Sum of the two values:"<<sum;

     return 0;
}
```

Output;-

```
C:\Users\USER\OneDrive\Desktop\c++> g++ lb3(3).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Sum of the two values:81
C:\Users\USER\OneDrive\Desktop\c++>
```

Question 4

Input;-

File Edit View

```
#include <iostream>
using namespace std;

int main() {
    int birthYear, currentYear = 2025;
    int age ;
    string name;
    cout<<"Enter your name:";
    getline (cin,name);

       cout<<"Enter your birth year:";
       cin>> birthYear;

       age = currentYear - birthYear;

       cout<<"Student Name:"<<name<<endl;
       cout<<"Student age:"<<age<<endl;

    return 0;
}
```

Output;-

```
C:\Users\USER\OneDrive\Desktop\c++> g++ lb3(4).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Enter your name:Yuwani
Enter your birth year:2001
Student Name:Yuwani
Student age:24

C:\Users\USER\OneDrive\Desktop\c++>
```

Question 5

Input;-

File Edit View

```
#include <iostream>
using namespace std;

int main() {
    int value, a=50,b =83;
      cout<<"before swap value of a:"<<a<<endl;
      cout<<"before swap value of b:"<<b<<endl;

        value = a;
        a=b;
        b= value;

      cout<<"after swap value of a:"<<a<<endl;
      cout<<"after swap value of b:"<<b<<endl;

    return 0;
}
```

Output;-

```
C:\Users\USER\OneDrive\Desktop\c++> g++ lb3(5).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
before swap value of a:50
before swap value of b:83
after swap value of a:83
after swap value of b:50
```

Question 6

Input;-

File Edit View

```
#include <iostream>
using namespace std;

int main() {
   int distance,time;
      cout<<"Distance(meters):";
      cin>>distance;

      cout<<"Time (second):";
      cin>>time;

   int speed= distance/time;
      cout<<"Average speed:"<<speed<<" m/s";

   return 0;
}
```

Output;-

```
C:\Users\USER\OneDrive\Desktop\c++> g++ lb3(6).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Distance(meters):560
Time (second):60
Average speed:9 m/s
C:\Users\USER\OneDrive\Desktop\c++>
```

The given example in the code , when 560m divide by 60s it won't give the correct answer 9.33 . Instead of that it only gives the answer 9. Because integer data type can't store decimal values.

To fix this problem we can use float data type or double data type which an store decimal values

```
File    Edit    View

#include <iostream>
using namespace std;

int main() {
   float averagespeed,distance,time;
      cout<<"Distance(meters):";
      cin>>distance;

      cout<<"Time (second):";
      cin>>time;

      averagespeed= distance/time;
      cout<<"Average speed:"<<averagespeed<<" m/s";

   return 0;
}
```

```
C:\Users\USER\OneDrive\Desktop\c++> g++ lb3(6)1.cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Distance(meters):560
Time (second):60
Average speed:9.33333 m/s
C:\Users\USER\OneDrive\Desktop\c++>
```

Question 7

Input;-

File Edit View

```
#include <iostream>
using namespace std;

int main() {
   float  f,c;
   cout<<"Enter tempertature (degrees Farenheit):";
   cin>>f;

   c=(5.0/9.0)*(f-32);
   cout<<"Tempertature reading (degrees celsius):"<<c<<endl;

   return 0;
}
```

Output;-

```
C:\Users\USER\OneDrive\Desktop\c++> g++ lb3(7).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Enter tempertature (degrees Farenheit):68
Tempertature reading (degrees celsius):20

C:\Users\USER\OneDrive\Desktop\c++>g++ lb3(7).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Enter tempertature (degrees Farenheit):150
Tempertature reading (degrees celsius):65.5556

C:\Users\USER\OneDrive\Desktop\c++>g++  lb3(7).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Enter tempertature (degrees Farenheit):212
Tempertature reading (degrees celsius):100

C:\Users\USER\OneDrive\Desktop\c++>g++  lb3(7).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Enter tempertature (degrees Farenheit):0
Tempertature reading (degrees celsius):-17.7778

C:\Users\USER\OneDrive\Desktop\c++>g++  lb3(7).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Enter tempertature (degrees Farenheit):-22
Tempertature reading (degrees celsius):-30

C:\Users\USER\OneDrive\Desktop\c++>g++  lb3(7).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Enter tempertature (degrees Farenheit):200
Tempertature reading (degrees celsius):93.3333

C:\Users\USER\OneDrive\Desktop\c++>g++ lb3(7).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Enter tempertature (degrees Farenheit):-200
Tempertature reading (degrees celsius):-128.889

C:\Users\USER\OneDrive\Desktop\c++>
```

Question 8

Input;-

File Edit View

```
#include <iostream>
using namespace std;

int main() {
    int num1=52, num2= 12;

     if(num1>num2){
        cout<<"Highest number:"<<num1<<endl;
     }
     else{
        cout<<"Highest number:"<<num2<<endl;
     }

     return 0;
 }
```

Output;-

```
C:\Users\USER\OneDrive\Desktop\c++> g++ lb3(8).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Highest number:52
```

Question 9

Input;-

File Edit View

```
#include <iostream>
using namespace std;

int main() {
    int Marks;
        cout<<"Enter marks:";
        cin>> Marks;

           if(Marks>=60){
               cout<<"Pass"<<endl;
           }
           else{
              cout<<"Fail"<<endl;
           }
           return 0;
}
```

Output;-

```
C:\Users\USER\OneDrive\Desktop\c++> g++ lb3(9).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Enter marks:56
Fail

C:\Users\USER\OneDrive\Desktop\c++>g++ lb3(9).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Enter marks:82
Pass
```

Question 10

Input;-

File Edit View

```
#include <iostream>
using namespace std;

int main() {
  string name;
  cout<<"Enter your name:";
  getline(cin,name);

  double basic,newSalary,increment;
  cout<<"Enter your basic salary:";
  cin>>basic;

      if(basic<5000){
           increment = basic*0.05;
      }
      else if (basic >= 5000 && basic < 10000){
            increment = basic*0.10;
      }
      else{
             increment = basic*0.15;
      }

      newSalary = basic+ increment;

       cout<<"Employee name:"<<name<<endl;
       cout<< "New Salary:"<<newSalary<<endl;

    return 0;
}
```

Output;-

```
C:\Users\USER\OneDrive\Desktop\c++>g++ lb3(10).cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Enter your name:Yuwani
Enter your basic salary:64500
Employee name:Yuwani
New Salary:74175
```